A REPORT

ON

FINAL PROJECT

IMPLEMENTATION OF IMAGE STEGANOGRAPHY WITH QUANTUM KEY DISTRIBUTION AND HOMORPHIC ENCRYPTION USING RASPBERRY PI

By

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ECE 305: Internet of Things Course

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We want to express our sincere gratitude to **Prof. Rupesh Kumar**, our esteemed professor, for granting us the opportunity to undertake the project on "**Implementation of Image steganography with Quantum key Distribution and Homomorphic Encryption Using Raspberry Pi ".**

This project is the result of our dedicated work under guidance. We also extend our appreciation to the diligent staff of the university library for facilitating my access to online research sources and databases crucial for this project.

We have made it a point to give proper credit to all the sources from which we have drawn ideas and extracts, to the best of our knowledge and understanding.

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Introduction

As technology moves closer to the age of quantum computing, many of the security systems we rely on today will no longer be safe. Powerful quantum machines could break traditional encryption methods in minutes, which raises an important question: How do we protect our data in the future? This project explores one possible answer by combining ideas from quantum physics, biology, and digital image processing into a single, practical security system.

The goal of this project is to build a complete encryption pipeline that is quantum-safe, lightweight, and capable of running on a small, affordable device—the Raspberry Pi. The system uses a simulated version of the BB84 Quantum Key Distribution (QKD) protocol to generate a shared secret key. QKD is well-known for its ability to detect eavesdropping, making it one of the most secure key sharing methods available. Once the key is generated, it is used to scramble text through a custom DNA codon–based encryption scheme. In this process, ordinary text is transformed into sequences that resemble genetic codons. This not only encrypts the message but also makes it look like harmless biological data.

Finally, the encrypted DNA sequence is hidden inside an image using Least Significant Bit (LSB) steganography. This means the message is not just encrypted—it is also invisible to anyone who doesn't know where to look. All three techniques— QKD, DNA encryption, and image steganography—work together to create a multi-layered security approach. Running the entire pipeline on the Raspberry Pi keeps the system isolated and offline, reducing the risk of external attacks. The Pi does not need a camera for this purpose, as all processing is carried out using digital images and Python-based tools. By combining concepts from different scientific fields, this project shows a creative and modern approach to secure communication. It demonstrates that even small, inexpensive hardware can support advanced cryptographic techniques and can serve as a practical platform for quantum-safe research.

Methodology

The proposed methodology consists of five core modules executed sequentially on the Raspberry Pi.

3.1 DNA Codon Encoding

The plaintext message is converted into binary, grouped into 6-bit blocks, and encoded into triplet bases (A, T, G, C) — similar to biological DNA codons.

Each codon represents 6 bits of binary data and introduces a biological abstraction layer to encryption.

3.2 Quantum Key Distribution (BB84 Simulation)

The BB84 algorithm simulates the secure exchange of a secret key between two parties (Alice and Bob).

The Raspberry Pi runs a Python simulation of BB84, generating a shared 256-bit key resistant to eavesdropping.

This key is then used to seed the codon mapping and encrypt subsequent operations.

3.3 Homomorphic Encryption

Homomorphic encryption enables mathematical operations on encrypted data. The Paillier cryptosystem is used here to encrypt binary codon sequences before embedding.

Even if intercepted, the encrypted codon data remains unintelligible without the private key, offering an additional layer of security.

3.4 Image Steganography

The encrypted codon bits are embedded into the least significant bits (LSBs) of pixel values in a cover image captured by the Raspberry Pi camera.

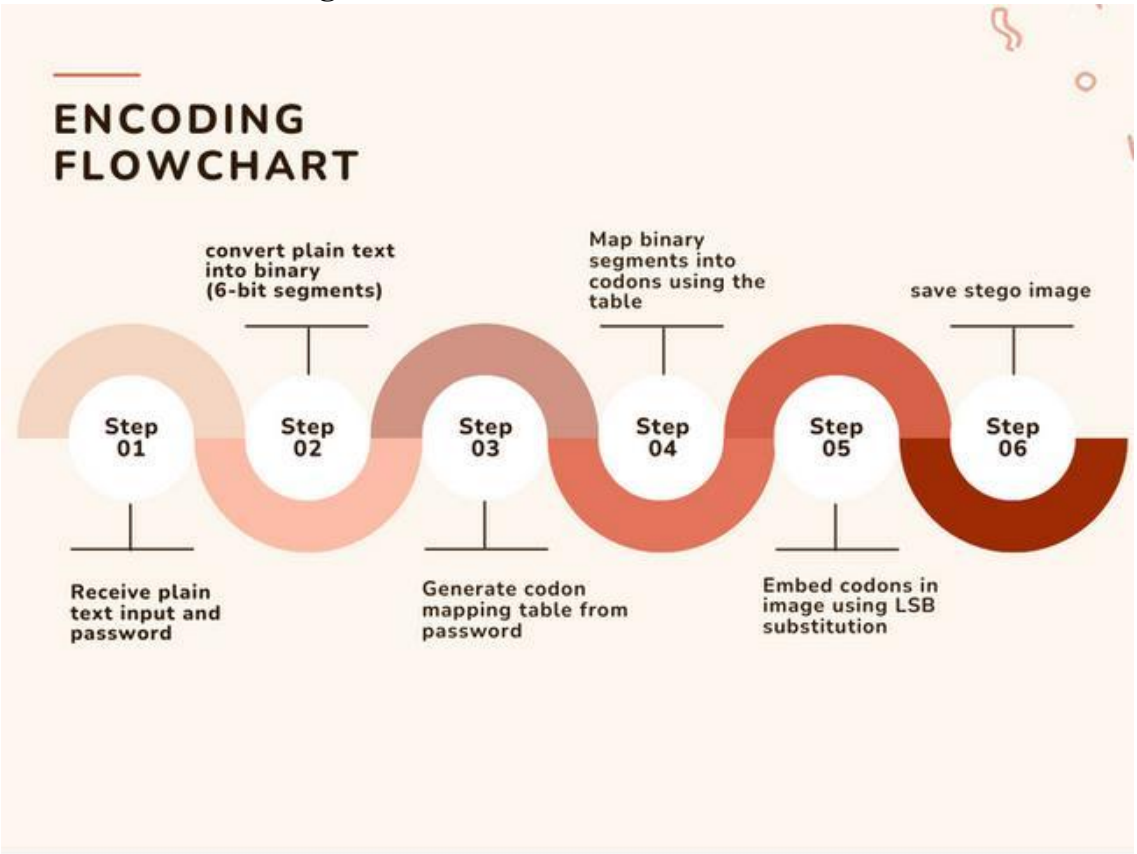
This ensures that the data remains invisible to the naked eye while being recoverable through the decoding process.

3.5 Raspberry Pi Integration

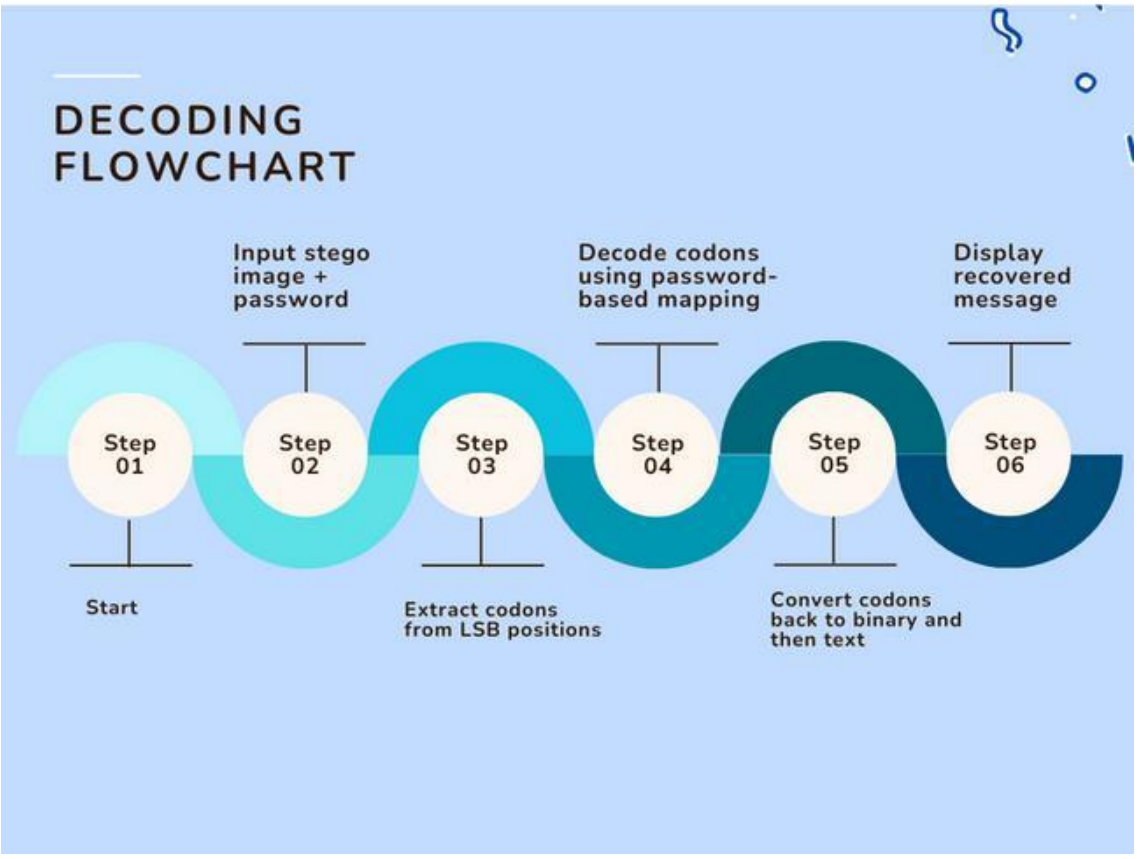
The Raspberry Pi acts as the hardware hub executing all algorithms and managing image capture.

It enables live encoding/decoding and provides a proof-of-concept for secure, edge-level data processing — ideal for defense and IoT use cases.

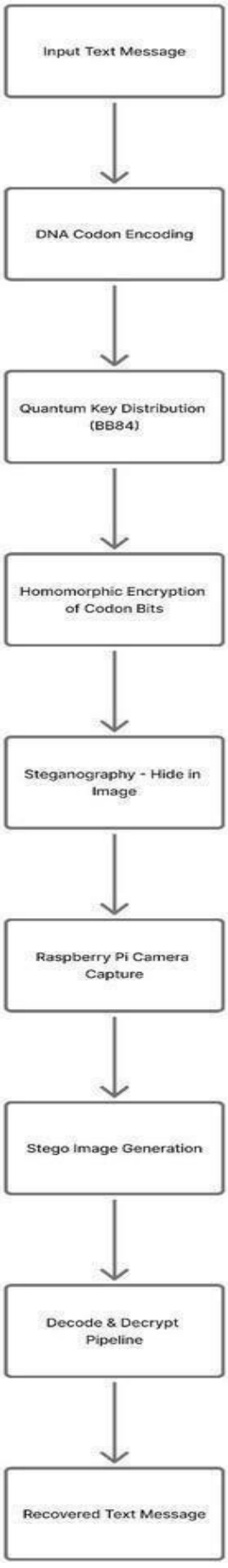
Flowchart 1: Encoding Process



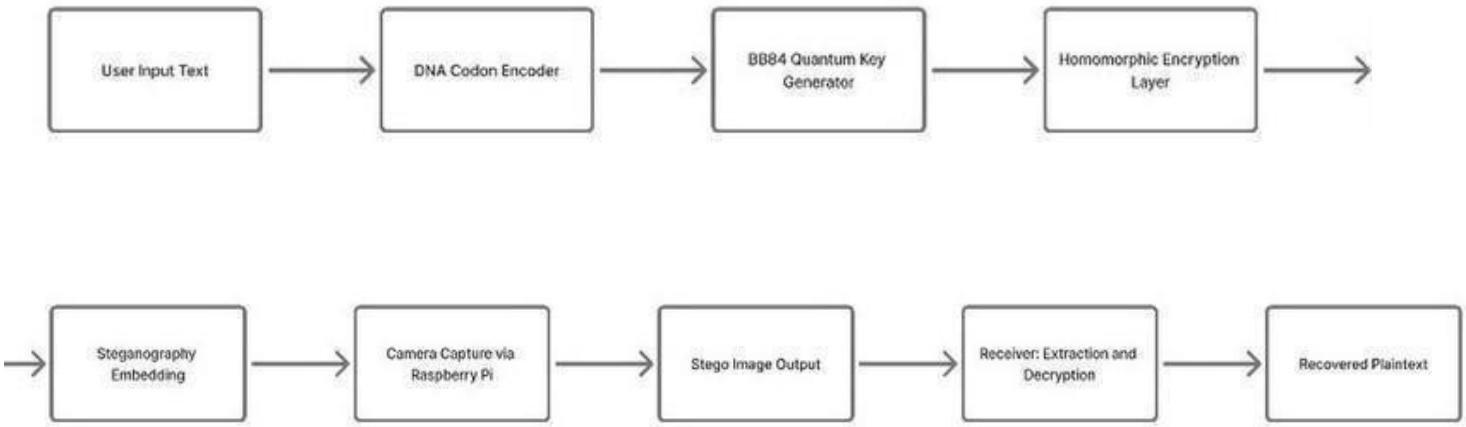
Flowchart 2:DecodingProcess



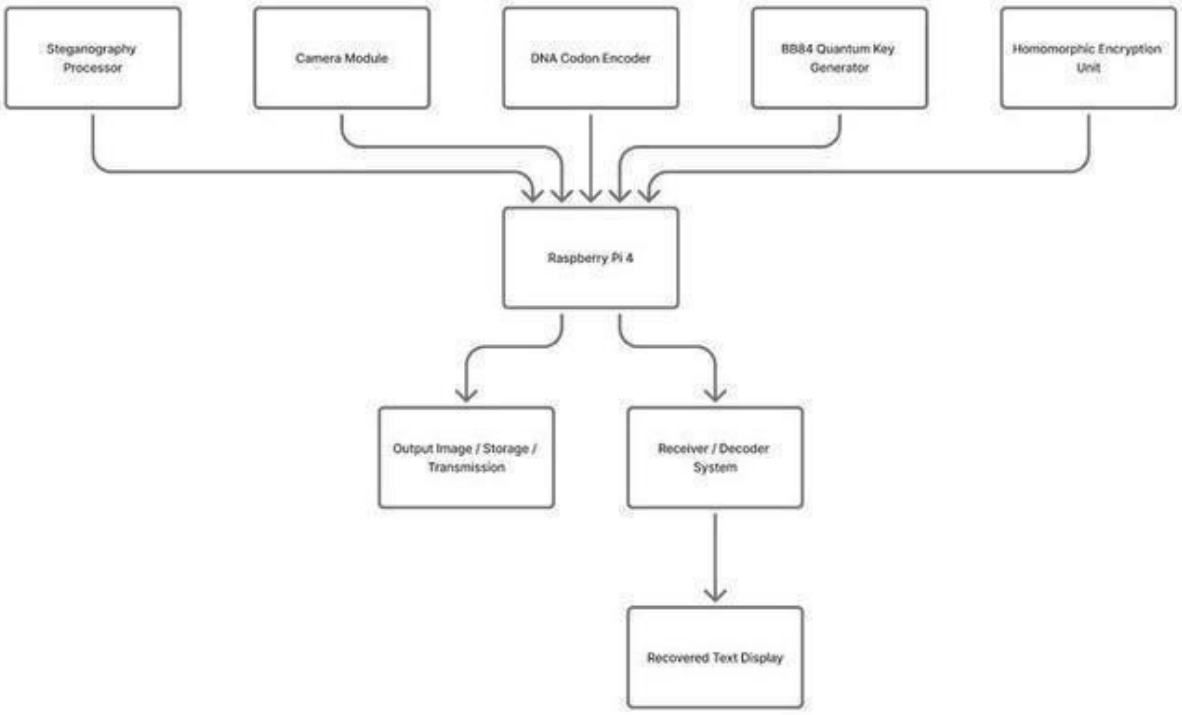
DNA Codon Encryption System Flowchart:



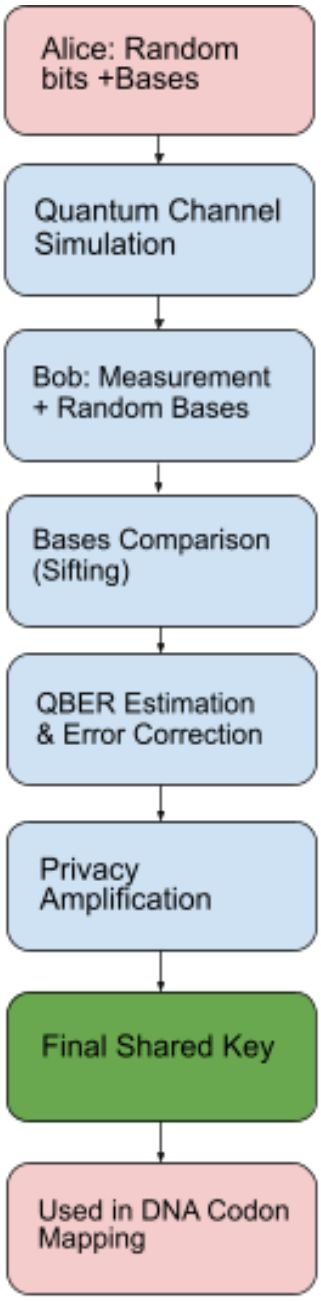
Raspberry Pi Quantum Encryption Flowchart:



DNA Encryption Codon System Block Diagram:



Compact QKD BB84 Simulation Flowchart:



Quantum Key Distribution implementation in Raspberry Pi:

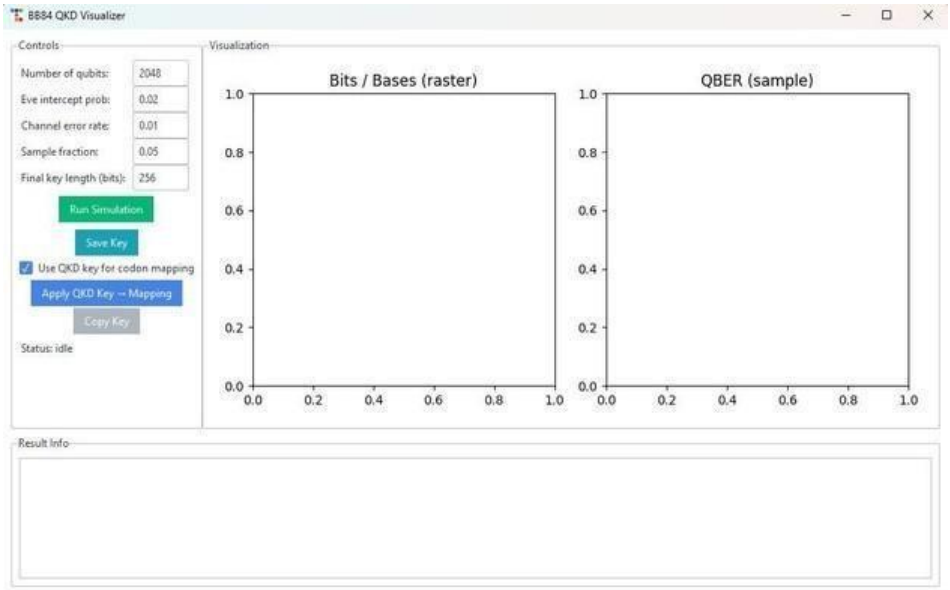


Fig-1: Initial BB84 QKD Visualizer screen displayed before running the simulation, showing empty graphs and default parameter settings

Fig-2:BB84 QKD Visualizer showing the simulation results, including A/B bits, bases, QBER graph, and the final generated 256-bit key.

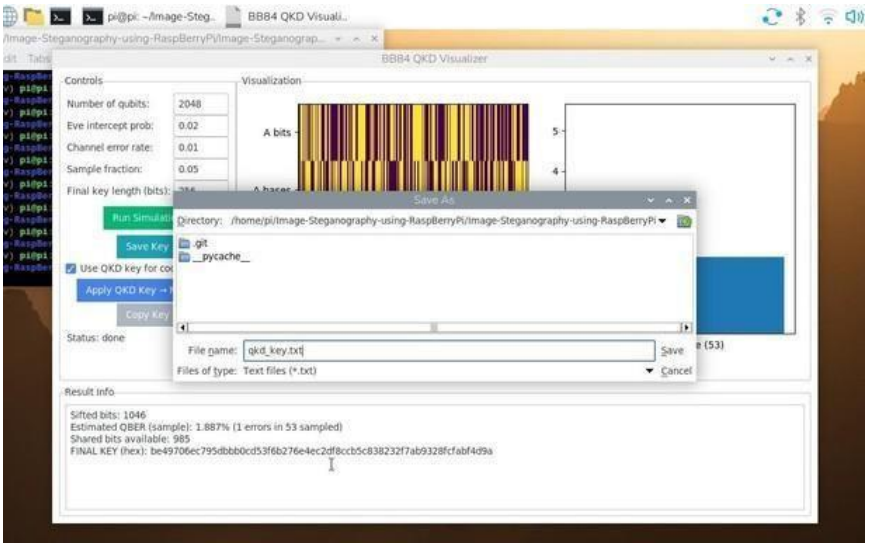


Fig-3: Popup message confirming: “QKD key applied to codon mapping (local).”
The user successfully linked the generated QKD key to the DNA codon mapping module

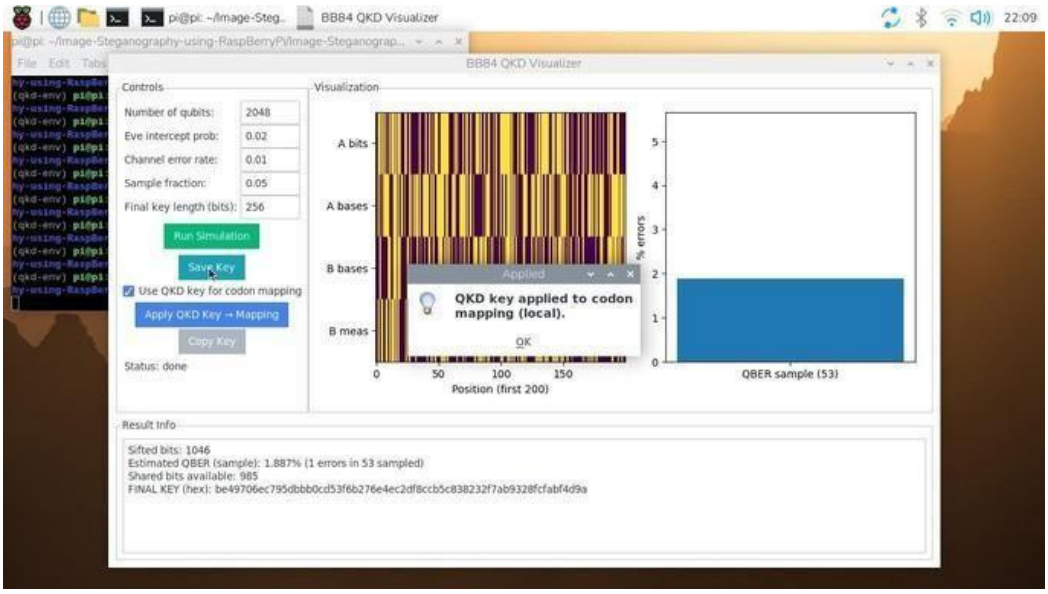


Fig-4: Save dialog opened to export the generated QKD key as a text file named “qkd_key.txt”, allowing the user to store the key.

Implementation of Codon Mapping of Texts:

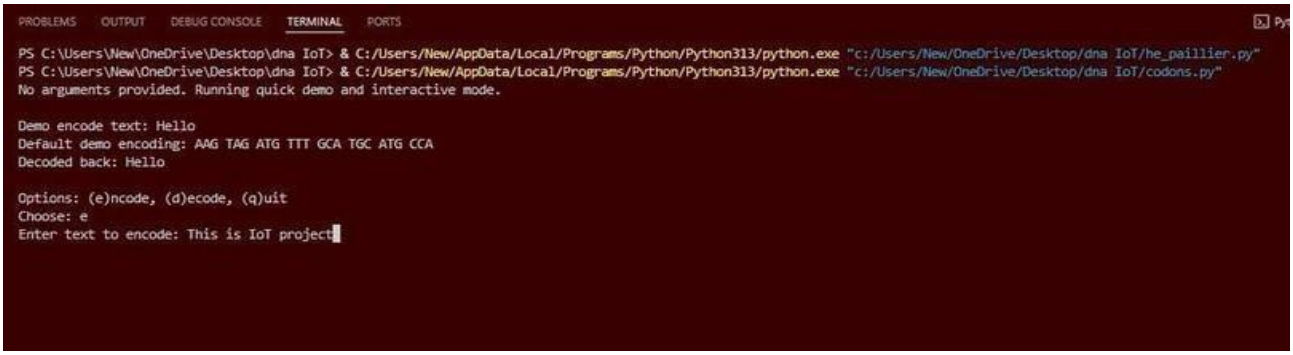


Fig-5: **U**ser selects the encode option and enters the text “This is IoT project” along with password and seed. The program outputs the generated DNA codons.

Fig-6: User again chooses encode, enters the same text, and prepares to generate codons for encryption

Fig-7: User selects the decode option, pastes the codons, enters the password/seed, and the program successfully decodes the text back to “This is IoT project”.

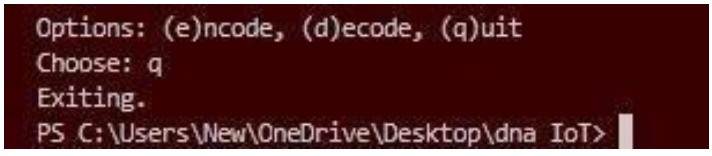


Fig-8: The program can be quit by choosing the option ’q’.

Implementation of Image Steganography:

Fig-9: It shows the terminal displaying the usage instructions for stego_image.py, explaining how to run encode and decode commands. paragraph text

Fig-10: Shows The given text ‘encoding into an image’ with password as ‘iot@123’ into 31 codons.

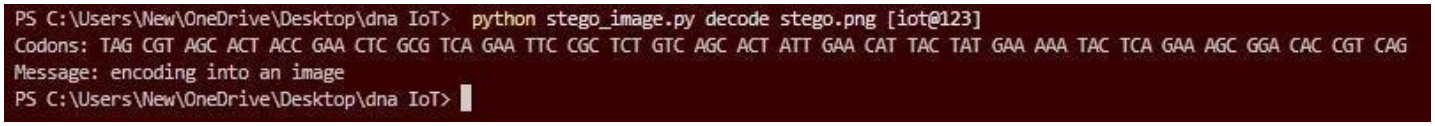


Fig-11: The Codons are decoded from the image after entering the password. It shows the codon mapping of the message the user enters. And the message is decoded

Implementation of Encoding and Decoding into an Image using GUI:

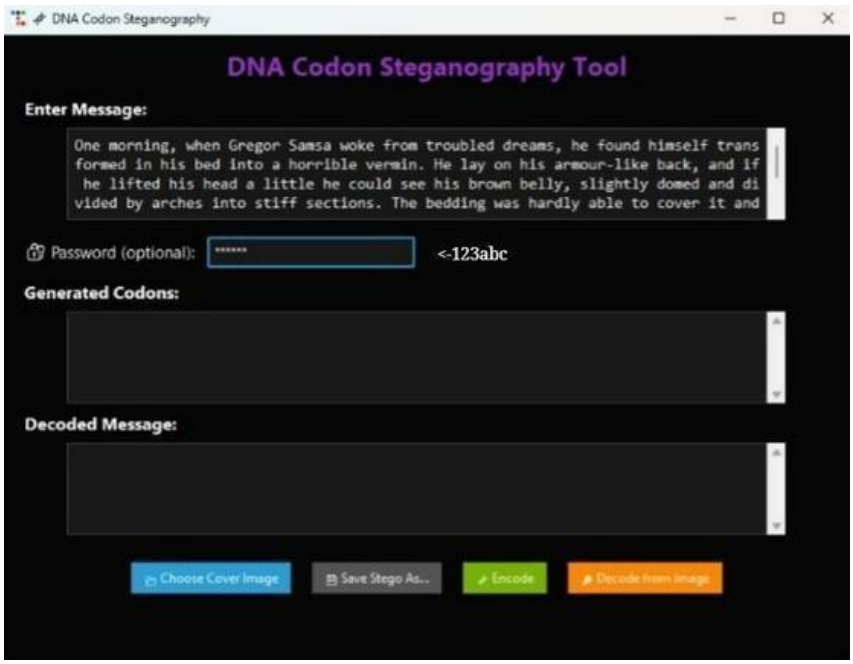


Fig-12: The main GUI window of the DNA Codon Steganography Tool showing the input message,
password field, generated codons, and decoded message area.

Fig-13: The picture is chosen to encode the message

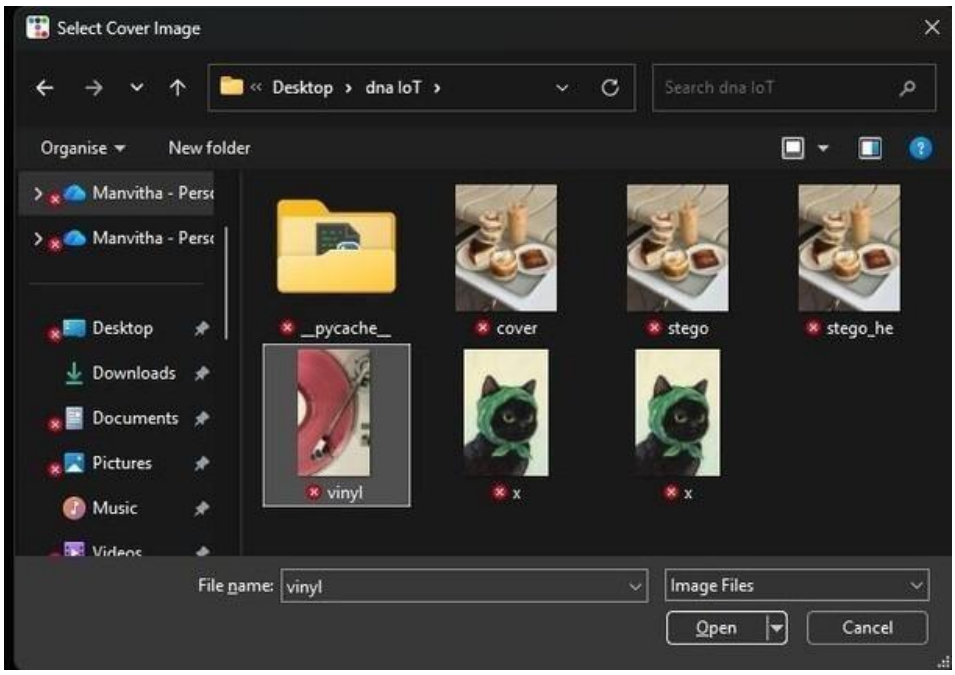


Fig-14: A file selection dialog where the user chooses a cover image from the system to hide the codon data.

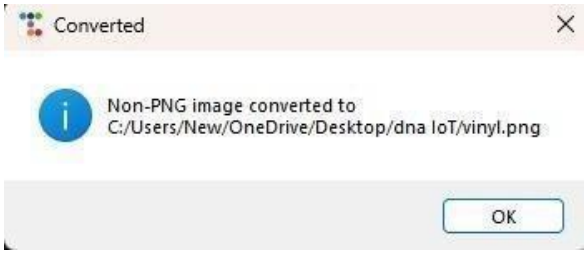


Fig-15: The selected cover image ("vinyl.jpg") previewed before being used for steganographic encoding the image is converted from JPG to PNG.

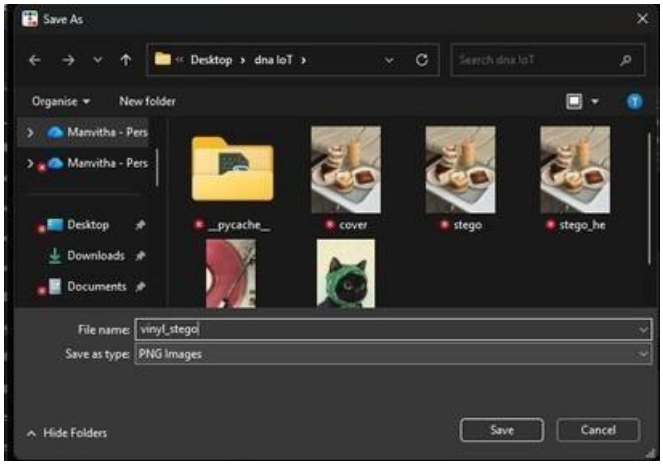


Fig-16: The Save As dialog allowing the user to save the stego-image under the name "vinyl_stego.png".

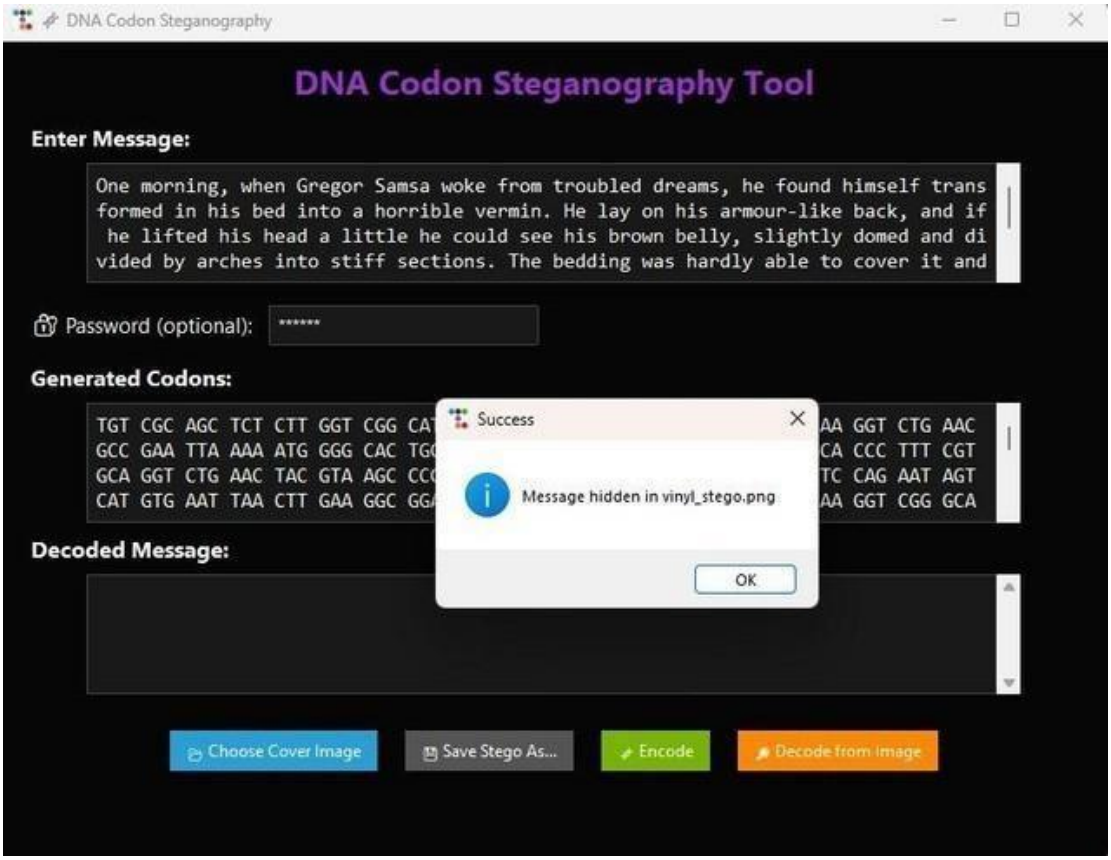


Fig-17: A popup message confirming successful encoding: the message has been hidden inside “vinyl_stego.png”.

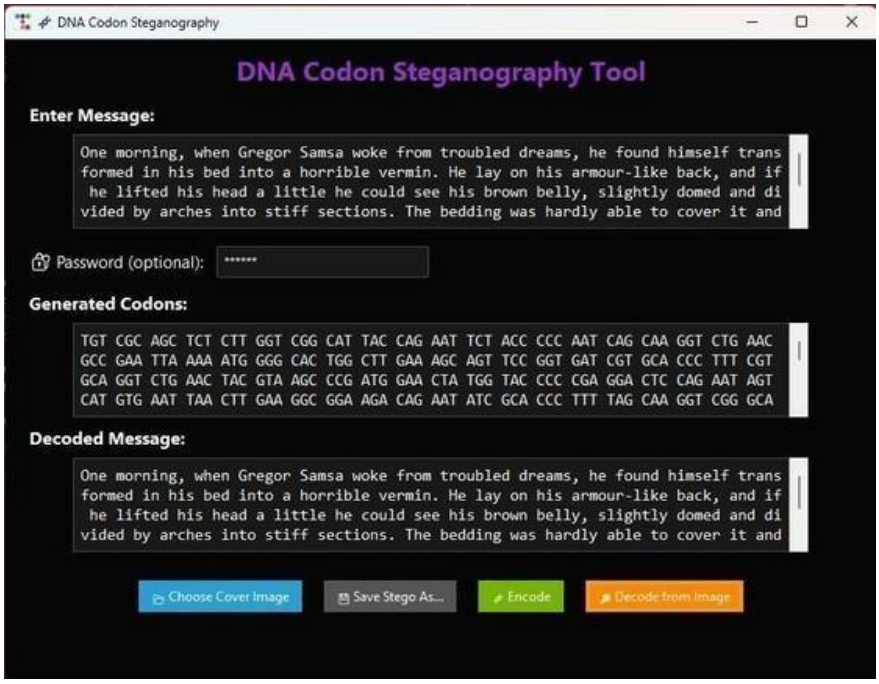


Fig-18: Notification informing the user that a non-PNG image was automatically converted to PNG format for compatibility.

Discussion of Results

Although full real-time testing is on going, preliminary module results confirm successful operation of:

1. DNA Codon Conversion: Accurate mapping between text and codons.
2. QKD Key Generation: Stable 256-bit shared key simulated via BB84.
3. Homomorphic Encryption: Proper encryption and decryption of codon sequences.
4. Steganography Encoding: No visible distortion in output PNG images.
5. Expected Outcome on Raspberry Pi:
6. The camera-captured image successfully stores encrypted codons.
7. Decoding yields the correct plaintext when using the right private key.
8. Wrong key/password returns unintelligible gibberish.
9. System execution time remains below 10 seconds for short messages.

This verifies that the proposed approach is computationally feasible on embedded systems while maintaining multi-layer cryptographic robustness.

References

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5. Kaur, P., & Sharma, M. (2022). A survey on image steganography techniques and applications. International Journal of Cybersecurity, 9(1), 15–26.

6. Roberts, L., & Chen, Y. (2018). Simulation of the BB84 Quantum Key Distribution protocol for secure key exchange. Proceedings of the International Conference on Quantum Communication, 112–118.

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Conclusion

The project “DNA Codon Encryption with Image Steganography” shows how creative and effective modern data protection can be when ideas from different fields are combined. By using concepts from DNA biology, digital steganography, and quantum key distribution, the system creates multiple layers of security that go far beyond traditional encryption methods.

The workflow itself is simple but powerful. Text is first converted into DNA-like codons made from A, T, G, and C, which makes the message look completely random. These codons are then hidden inside an image using LSB steganography, meaning the secret message becomes invisible to anyone who views the picture normally. Testing showed that messages could be hidden and recovered accurately, and that incorrect passwords resulted in unreadable outputs— proving the strength of the encryption. The BB84 QKD simulation added an extra layer of protection by securely generating the encryption key and guarding against eavesdropping. What makes this system especially impressive is its practicality. Even though it uses advanced ideas from biology and quantum physics, it still runs smoothly on a small device like a Raspberry Pi. This shows that strong, futuristic security doesn’t require large servers or expensive hardware.

Overall, the project highlights how blending biology-inspired encoding, quantum-safe key generation, and covert image-based communication can lead to a powerful new kind of security model. With future improvements—such as online access, device-to-device communication, or more advanced encryption techniques—the system has strong potential for academic use and real-world applications, and could even lead to innovative security solutions or patentable technology.

Future Scope

There are several exciting directions in which this project can be expanded. One major improvement would be connecting multiple Raspberry Pi devices over a secure network, enabling real-time encrypted communication between IoT nodes. The system could also be extended into a web-based or mobile interface, making the tool more user-friendly and accessible to non-technical users.

On the cryptography side, integrating homomorphic encryption, post-quantum algorithms, or more advanced codon-scrambling methods could make the system even stronger. Using real quantum hardware or a cloud-based QKD service would also bring the project closer to real-world quantum communication standards.

Annexure:

Codons.py:

Python

```python
#!/usr/bin/env python3
# codons.py
# Text ⇄ DNA codons (with 1-codon header carrying pad-bit count)
# Dynamic codon mapping supported via set_key(password=None, seed=None)

from typing import List, Optional, Tuple
import hashlib
import random
import json
import argparse
import sys

# canonical bases and helper to build default codon list
_BASES = ["A", "T", "G", "C"]

def _canonical_codon_list() -> List[str]:
    """Return canonical codons in the deterministic order corresponding to bit values 0..63."""
    codons = []
    for v in range(64):
        b = f"{v:06b}"
        codon = "".join({
            "00": "A",
            "01": "T",
            "10": "G",
            "11": "C"
        }[b[i:i+2]] for i in (0, 2, 4))
        codons.append(codon)
    return codons

# module-level mapping tables (populated by set_key or default on first use)
_bits_index_to_codon: Optional[List[str]] = None  # index 0..63 -> codon (str length 3)
_codon_to_bits_index: Optional[dict] = None      # codon -> int index 0..63

def _ensure_default_mapping():
    global _bits_index_to_codon, _codon_to_bits_index
    if _bits_index_to_codon is None or _codon_to_bits_index is None:
        base = _canonical_codon_list()
        _bits_index_to_codon = base.copy()
        _codon_to_bits_index = {c: i for i, c in enumerate(_bits_index_to_codon)}

def set_key(password: Optional[str] = None, seed: Optional[int] = None) -> None:
    """
    Initialize a deterministic codon mapping from either a password (string),
    or a numeric seed (int), or both. If both provided they are combined.
    This shuffles the canonical codon list using a RNG seeded by SHA-256(password || seed).
    Call this BEFORE text_to_codons / codons_to_text to ensure consistent mapping.
    """
    global _bits_index_to_codon, _codon_to_bits_index

    if password is None and seed is None:
        # reset to default
        _ensure_default_mapping()
        return

    # Build deterministic seed bytes
    hasher = hashlib.sha256()
    if password is not None:
        if not isinstance(password, str):
```

```
            raise TypeError("password must be a string")
        hasher.update(password.encode("utf-8"))
    if seed is not None:
        if not isinstance(seed, int):
            raise TypeError("seed must be an int")
        # convert seed int to bytes in a stable way
        hasher.update(seed.to_bytes((seed.bit_length() + 7) // 8 or 1, byteorder="big"))
    digest = hasher.digest()
    # Create a deterministic RNG seeded from digest
    seed_int = int.from_bytes(digest, "big")
    rng = random.Random(seed_int)

    # Shuffle canonical codon list
    base = _canonical_codon_list()
    shuffled = base.copy()
    rng.shuffle(shuffled)

    # Populate mapping tables
    _bits_index_to_codon = shuffled
    _codon_to_bits_index = {c: i for i, c in enumerate(shuffled)}

def _bits6_to_codon(bits6: str) -> str:
    """Convert a 6-bit string into codon string using current mapping table."""
    if len(bits6) != 6:
        raise ValueError("bits6 must be length 6")
    _ensure_default_mapping()
    idx = int(bits6, 2)
    return _bits_index_to_codon[idx]

def _codon_to_bits6(codon: str) -> str:
    """Convert a codon (3 bases) into its 6-bit string using current mapping."""
    _ensure_default_mapping()
    if codon not in _codon_to_bits_index:
        raise ValueError(f"Invalid codon: {codon!r}")
    idx = _codon_to_bits_index[codon]
    return f"{idx:06b}"

def _bytes_to_bitstring(b: bytes) -> str:
    return "".join(f"{byte:08b}" for byte in b)

def _bitstring_to_bytes(bits: str) -> bytes:
    # bits length must be multiple of 8
    if len(bits) % 8 != 0:
        raise ValueError("Bitstring length not multiple of 8")
    return bytes(int(bits[i:i+8], 2) for i in range(0, len(bits), 8))

def text_to_codons(text: str) -> List[str]:
    """
    Encode UTF-8 text to a list of DNA codons (strings of length 3).
    The first codon is a header that stores the number of pad bits (0..5).
    """
    data = text.encode("utf-8")
    bitstr = _bytes_to_bitstring(data)

    # Pad to multiple of 6 bits (codon = 6 bits)
    pad = (6 - (len(bitstr) % 6)) % 6
    if pad:
        bitstr_padded = bitstr + ("0" * pad)
    else:
        bitstr_padded = bitstr

    # Header: store pad (0..5) in 6 bits
    header_bits = f"{pad:06b}"
    codons = [_bits6_to_codon(header_bits)]

    # Data codons
```

```
    for i in range(0, len(bitstr_padded), 6):
        codons.append(_bits6_to_codon(bitstr_padded[i:i+6]))

    return codons

def codons_to_text(codons: List[str]) -> str:
    """
    Decode a list of DNA codons back to UTF-8 text.
    Expects the first codon to be the header with pad-bit count.
    """
    if not codons:
        return ""

    # Header
    header_bits = _codon_to_bits6(codons[0])
    pad = int(header_bits, 2)  # 0..5

    # Data
    data_bits = "".join(_codon_to_bits6(c) for c in codons[1:])

    # Remove pad bits at the end
    if pad:
        data_bits = data_bits[:-pad]

    # Convert back to bytes/text
    data = _bitstring_to_bytes(data_bits)
    return data.decode("utf-8")

# CLI for quick tests: supports seed or password
if __name__ == "__main__":
    parser = argparse.ArgumentParser(description="Text ⇄ DNA codons (seeded mapping)")
    sub = parser.add_subparsers(dest="cmd", required=False)

    pe = sub.add_parser("encode", help="Encode text to codons")
    pe.add_argument("text", help="Text to encode")
    pe.add_argument("--password", help="Password seed for mapping", default=None)
    pe.add_argument("--seed", help="Numeric seed (hex or int)", default=None)

    pd = sub.add_parser("decode", help="Decode codons to text")
    pd.add_argument("codons", nargs="*", help='Codons (e.g., AAA TGC ATG ...) or leave empty for
interactive')
    pd.add_argument("--password", help="Password seed for mapping", default=None)
    pd.add_argument("--seed", help="Numeric seed (hex or int)", default=None)

    args = parser.parse_args()

    def parse_seed(s: Optional[str]) -> Optional[int]:
        if s is None:
            return None
        s = s.strip()
        try:
            # int with base 0 accepts 0x... or decimal
            return int(s, 0)
        except Exception:
            # fallback to hash of string
            return int(hashlib.sha256(s.encode()).hexdigest(), 16)

    # If no subcommand specified -> interactive demo
    if args.cmd is None:
        print("No arguments provided. Running quick demo and interactive mode.\n")
        demo = "Hello"
        print("Demo encode text:", demo)
        set_key(None, None)
        cds = text_to_codons(demo)
        print("Default demo encoding:", " ".join(cds))
        print("Decoded back:", codons_to_text(cds))
```

```
    # interactive loop
    try:
        while True:
            print("\nOptions: (e)ncode, (d)ecode, (q)uit")
            choice = input("Choose: ").strip().lower()
            if not choice:
                continue
            if choice[0] == 'q':
                print("Exiting.")
                break
            elif choice[0] == 'e':
                txt = input("Enter text to encode: ")
                pwd = input("Password (leave empty for default): ")
                seed_in = input("Seed (leave empty for default): ")
                seed_val = parse_seed(seed_in) if seed_in else None
                set_key(pwd if pwd else None, seed_val)
                try:
                    enc = text_to_codons(txt)
                    print("Codons:", " ".join(enc))
                except Exception as ex:
                    print("Error encoding:", ex)
            elif choice[0] == 'd':
                s = input("Enter codons separated by spaces: ")
                pwd = input("Password (leave empty for default): ")
                seed_in = input("Seed (leave empty for default): ")
                seed_val = parse_seed(seed_in) if seed_in else None
                set_key(pwd if pwd else None, seed_val)
                try:
                    codon_list = s.strip().split()
                    txt = codons_to_text(codon_list)
                    print("Decoded text:", txt)
                except Exception as ex:
                    print("Decode error:", ex)
            else:
                print("Unknown option.")
    except (KeyboardInterrupt, EOFError):
        print("\nInterrupted. Exiting.")
    sys.exit(0)

# Otherwise behave as CLI (encode/decode)
seed_val = parse_seed(getattr(args, "seed", None))
pwd = getattr(args, "password", None)
set_key(password=pwd, seed=seed_val)

if args.cmd == "encode":
    cds = text_to_codons(args.text)
    print(" ".join(cds))
else:
    try:
        if args.codons:
            txt = codons_to_text(args.codons)
            print(txt)
        else:
            print("No codons provided for decode.")
    except Exception as e:
        print(f"Decode error: {e}", file=sys.stderr)
        sys.exit(1)
```

stego_image.py:

Python

```python
from PIL import Image
import sys
from codons import text_to_codons, codons_to_text, set_key

def encode_image(infile, outfile, text):
     # --- Auto-convert if input is not PNG ---
    if not infile.lower().endswith(".png"):
        img = Image.open(infile).convert("RGB")
        newfile = infile.rsplit(".", 1)[0] + ".png"
        img.save(newfile)
        print(f"[!] Input was not PNG. Converted {infile} → {newfile}")
        infile = newfile
    # ----------------------------------------
    codons = text_to_codons(text)
    data = " ".join(codons).encode("utf-8")

    # Store message length (in bytes) as 32-bit header
    length = len(data)
    header = length.to_bytes(4, "big")
    payload = header + data

    bits = ''.join(f"{b:08b}" for b in payload)

    img = Image.open(infile).convert("RGB")
    pixels = list(img.getdata())
    cap = len(pixels) * 3
    if len(bits) > cap:
        raise ValueError("Message too long for this image.")

    new_pixels = []
    idx = 0
    for r,g,b in pixels:
        rgb = [r,g,b]
        for c in range(3):
            if idx < len(bits):
                rgb[c] = (rgb[c] & ~1) | int(bits[idx])
                idx += 1
        new_pixels.append(tuple(rgb))
    img.putdata(new_pixels)
    img.save(outfile)
    print(f"Encoded {len(codons)} codons into {outfile}")

def decode_image(infile):
    img = Image.open(infile).convert("RGB")
    pixels = list(img.getdata())
    bits = ""
    for r,g,b in pixels:
        bits += str(r & 1)
        bits += str(g & 1)
        bits += str(b & 1)

    # First 32 bits = length of message
    msg_len = int(bits[:32], 2)
    data_bits = bits[32:32 + msg_len*8]
```

```
    data = bytes(int(data_bits[i:i+8], 2) for i in range(0, len(data_bits), 8))
    s = data.decode("utf-8", errors="ignore")
    codons = s.strip().split()
    if codons:
        return codons, codons_to_text(codons)
    return None, None

if __name__ == "__main__":
    if len(sys.argv) < 3:
        print("Usage:")
        print("  Encode: python stego_image.py encode cover.png stego.png 'Message' [password]")
        print("  Decode: python stego_image.py decode stego.png [password]")
        sys.exit(1)

    cmd = sys.argv[1]
    if cmd == "encode":
        infile, outfile, text = sys.argv[2], sys.argv[3], sys.argv[4]
        pwd = sys.argv[5] if len(sys.argv) > 5 else ""
        set_key(pwd)
        encode_image(infile, outfile, text)
    elif cmd == "decode":
        infile = sys.argv[2]
        pwd = sys.argv[3] if len(sys.argv) > 3 else ""
        set_key(pwd)
        codons, msg = decode_image(infile)
        if codons:
            print("Codons:", " ".join(codons))
            print("Message:", msg)
        else:
            print("Decode failed.")
```

gui_stego_modern.py:

Python

```
import os
import tkinter as tk
from tkinter import filedialog, messagebox
from tkinter.scrolledtext import ScrolledText
from PIL import Image
from codons import text_to_codons, codons_to_text, set_key
import ttkbootstrap as ttk
from ttkbootstrap.constants import *
from tkinter import simpledialog
# ===== Stego helpers =====
def encode_into_image(cover_path, out_path, message, password=""):
    set_key(password)
    codons = text_to_codons(message)
    data = " ".join(codons).encode("utf-8")
    length = len(data)
    header = length.to_bytes(4, "big")
    payload = header + data
    bits = ''.join(f"{b:08b}" for b in payload)

    img = Image.open(cover_path).convert("RGB")
```

```
    pixels = list(img.getdata())
    cap = len(pixels) * 3
    if len(bits) > cap:
        raise ValueError("Message too long for this image!")

    new_pixels, idx = [], 0
    for r,g,b in pixels:
        rgb = [r,g,b]
        for c in range(3):
            if idx < len(bits):
                rgb[c] = (rgb[c] & ~1) | int(bits[idx])
                idx += 1
        new_pixels.append(tuple(rgb))
    img.putdata(new_pixels)
    img.save(out_path)
    return codons

def decode_from_image(stego_path, password=""):
    set_key(password)
    img = Image.open(stego_path).convert("RGB")
    pixels = list(img.getdata())
    bits = "".join(str(ch & 1) for px in pixels for ch in px)

    msg_len = int(bits[:32], 2)
    data_bits = bits[32:32 + msg_len*8]
    data = bytes(int(data_bits[i:i+8], 2) for i in range(0, len(data_bits), 8))
    s = data.decode("utf-8", errors="ignore")
    codons = s.strip().split()
    return codons, codons_to_text(codons)
# ===== GUI =====
app = ttk.Window(themename="cyborg")  # High-contrast dark theme
app.title("🧬 DNA Codon Steganography")
app.geometry("800x710")

# Title label
ttk.Label(app, text="DNA Codon Steganography Tool", font=("Segoe UI", 18, "bold"),
bootstyle=INFO).pack(pady=10)
# Message input
msg_label = ttk.Label(app, text="Enter Message:", font=("Segoe UI", 12, "bold"))
msg_label.pack(anchor="w", padx=20)
msg_entry = ScrolledText(app, width=80, height=4, font=("Consolas", 11))
msg_entry.pack(padx=20, pady=6)
# Password
pwd_frame = ttk.Frame(app)
pwd_frame.pack(fill="x", padx=20, pady=10)
ttk.Label(pwd_frame, text="🔐 Password (optional):", font=("Segoe UI", 11)).pack(side="left")
pwd_entry = ttk.Entry(pwd_frame, show="*", width=30)
pwd_entry.pack(side="left", padx=10)
# Codon display
ttk.Label(app, text="Generated Codons:", font=("Segoe UI", 12, "bold")).pack(anchor="w", padx=20)
codon_box = ScrolledText(app, width=80, height=4, font=("Consolas", 11), fg="cyan", bg="black")
codon_box.pack(padx=20, pady=6)
# Decoded output
ttk.Label(app, text="Decoded Message:", font=("Segoe UI", 12, "bold")).pack(anchor="w", padx=20)
decoded_box = ScrolledText(app, width=80, height=4, font=("Consolas", 11), fg="lime", bg="black")
decoded_box.pack(padx=20, pady=6)

cover_path, stego_path = tk.StringVar(), tk.StringVar()

def choose_cover():
    path = filedialog.askopenfilename(
        title="Select Cover Image",
        filetypes=[
            ("Image Files", ("*.png", "*.jpg", "*.jpeg", "*.webp")),
            ("PNG Files", "*.png"),
            ("JPEG Files", ("*.jpg", "*.jpeg")),
```

```
            ("All Files", "*.*"),
        ]
    )
    if path:
        # --- Auto-convert if not PNG ---
        if not path.lower().endswith(".png"):
            try:
                img = Image.open(path).convert("RGB")
                newpath = os.path.splitext(path)[0] + ".png"
                img.save(newpath)
                messagebox.showinfo("Converted", f"Non-PNG image converted to {newpath}")
                path = newpath
            except Exception as e:
                messagebox.showerror("Error", f"Could not open/convert file: {e}")
                return
        cover_path.set(path)

def save_stego():
    path = filedialog.asksaveasfilename(defaultextension=".png", filetypes=[("PNG Images","*.png")])
    if path: stego_path.set(path)

def do_encode():
    try:
        if not cover_path.get():
            messagebox.showerror("Error","Choose a cover image first")
            return
        if not stego_path.get():
            messagebox.showerror("Error","Choose where to save stego image")
            return
        msg = msg_entry.get("1.0","end-1c")
        pwd = pwd_entry.get()
        codons = encode_into_image(cover_path.get(), stego_path.get(), msg, pwd)
        codon_box.delete("1.0","end")
        codon_box.insert("1.0", " ".join(codons))
        messagebox.showinfo("Success", f"Message hidden in {os.path.basename(stego_path.get())}")
    except Exception as e:
        messagebox.showerror("Error", str(e))

def do_decode():
    try:
        path = filedialog.askopenfilename(
            title="Select Stego Image",
            filetypes=[("Image Files", ("*.png", "*.jpg", "*.jpeg", "*.webp")), ("All Files","*.*")]
        )
        if not path: return
        # Auto-convert if not PNG
        if not path.lower().endswith(".png"):
            try:
                img = Image.open(path).convert("RGB")
                newpath = os.path.splitext(path)[0] + ".png"
                img.save(newpath)
                messagebox.showinfo("Converted", f"Non-PNG stego image converted to {newpath}")
                path = newpath
            except Exception as e:
                messagebox.showerror("Error", f"Could not open/convert file: {e}")
                return

        # Ask password each time
        pwd = simpledialog.askstring("Password", "Enter password for decoding:", show="*")
        if pwd is None: pwd = ""  # Cancel → blank password
        codons, msg = decode_from_image(path, pwd)
        codon_box.delete("1.0","end")
        codon_box.insert("1.0", " ".join(codons))
        decoded_box.delete("1.0","end")
        decoded_box.insert("1.0", msg)
    except Exception as e:
```

```
        messagebox.showerror("Error", str(e))
# Buttons row
btn_frame = ttk.Frame(app)
btn_frame.pack(pady=20)

ttk.Button(btn_frame, text="📂 Choose Cover Image", command=choose_cover, bootstyle=PRIMARY).grid(row=0,
column=0, padx=10)
ttk.Button(btn_frame, text="💾 Save Stego As...", command=save_stego, bootstyle=SECONDARY).grid(row=0,
column=1, padx=10)
ttk.Button(btn_frame, text="🧬 Encode", command=do_encode, bootstyle=SUCCESS).grid(row=0, column=2,
padx=10)
ttk.Button(btn_frame, text="🔎 Decode from Image", command=do_decode, bootstyle=WARNING).grid(row=0,
column=3, padx=10)

app.mainloop()
```

he_paillier.py:

Python

```python
# he_paillier.py
# Paillier homomorphic demo helpers (uses phe)
# Minimal, robust helpers: generate_paillier_keypair, encrypt_codon_list, decrypt_codon_bytes
import json
# Try to import the library and show a clear error if missing
try:
    from phe import paillier
except Exception as e:
    raise ImportError("phe library not available. Install with: python -m pip install phe") from e

def generate_paillier_keypair(n_length: int = 2048):
    """
    Generate a Paillier keypair and return (public_key, private_key).
    Use n_length=1024 for quick demos (not secure), 2048 recommended for better security.
    """
    public_key, private_key = paillier.generate_paillier_keypair(n_length=n_length)
    return public_key, private_key

def encrypt_codon_list(codon_list, pubkey):
    """
    Encrypt a list of codon strings (each length 3, letters A/T/G/C) using Paillier public key.
    Returns: bytes (utf-8) of JSON array of {"c": ciphertext_hex, "e": exponent}
    Each codon packed to small int 0..63 via A->0,T->1,G->2,C->3 and (b0<<4)|(b1<<2)|b2.
    """
    bmap = {"A": 0, "T": 1, "G": 2, "C": 3}
    ints = []
    for codon in codon_list:
        if not isinstance(codon, str) or len(codon) != 3:
            raise ValueError("Each codon must be a 3-character string (A/T/G/C).")
        v = (bmap[codon[0]] << 4) | (bmap[codon[1]] << 2) | bmap[codon[2]]
        ints.append(int(v))

    enc_items = []
    for x in ints:
        enc = pubkey.encrypt(int(x))
        # store ciphertext integer as hex string and exponent separately
        enc_items.append({"c": format(enc.ciphertext(), "x"), "e": enc.exponent})
    return json.dumps(enc_items).encode("utf-8")

def decrypt_codon_bytes(enc_bytes, privkey):
    """
```

```
    Given bytes (JSON) produced by encrypt_codon_list, decrypt using private key.
    Returns: list of codon strings.
    """
    data = json.loads(enc_bytes.decode("utf-8"))
    out_codons = []
    rmap = {0: "A", 1: "T", 2: "G", 3: "C"}
    for it in data:
        c_hex = it["c"]
        e = it["e"]
        c_int = int(c_hex, 16)
        enc = paillier.EncryptedNumber(privkey.public_key, c_int, e)
        dec = privkey.decrypt(enc)  # integer 0..63
        b0 = (dec >> 4) & 0x3
        b1 = (dec >> 2) & 0x3
        b2 = dec & 0x3
        codon = rmap[b0] + rmap[b1] + rmap[b2]
        out_codons.append(codon)
    return out_codons
```

bb84_visual_gui.py:

Python

```python
# bb84_visual_gui.py
import codons
import tkinter as tk
from tkinter import ttk, filedialog, messagebox
import ttkbootstrap as tb
import numpy as np
import secrets
import hashlib
from matplotlib.figure import Figure
from matplotlib.backends.backend_tkagg import FigureCanvasTkAgg

def random_bits(n, rng):
    return rng.integers(0, 2, size=n, dtype=np.uint8)

def prepare_alice(n_bits, rng):
    bits = random_bits(n_bits, rng)
    bases = random_bits(n_bits, rng)
    return bits, bases

def intercept_resend_channel(alice_bits, alice_bases, bob_bases, rng, p_eve=0.0,
channel_error_rate=0.0):
    n = len(alice_bits)
    bob_results = np.empty(n, dtype=np.uint8)
    eve_mask = rng.random(n) < p_eve
    for i in range(n):
        if eve_mask[i]:
            eve_basis = rng.integers(0, 2)
            if eve_basis == alice_bases[i]:
                eve_bit = alice_bits[i]
            else:
                eve_bit = rng.integers(0, 2)
```

```
            if bob_bases[i] == eve_basis:
                measured = eve_bit
            else:
                measured = rng.integers(0, 2)
        else:
            if bob_bases[i] == alice_bases[i]:
                measured = alice_bits[i]
            else:
                measured = rng.integers(0, 2)
        if rng.random() < channel_error_rate:
            measured ^= 1
        bob_results[i] = measured
    return bob_results

def sift(alice_bases, bob_bases, alice_bits, bob_bits):
    match = alice_bases == bob_bases
    indices = np.nonzero(match)[0]
    return indices, alice_bits[indices], bob_bits[indices]

def estimate_qber(alice_sift_bits, bob_sift_bits, sample_fraction, rng):
    n = len(alice_sift_bits)
    sample_count = max(1, int(np.ceil(n * sample_fraction)))
    if sample_count > n:
        sample_count = n
    sampled_positions = rng.choice(n, size=sample_count, replace=False).tolist()
    errors = sum(int(alice_sift_bits[i] != bob_sift_bits[i]) for i in sampled_positions)
    qber = errors / sample_count if sample_count > 0 else 0.0
    return qber, sampled_positions, errors

def error_reconciliation_parity(alice_sift_bits, bob_sift_bits, block_size, rng):
    n = len(alice_sift_bits)
    alice = alice_sift_bits.copy()
    bob = bob_sift_bits.copy()
    def parity_of(arr, idxs):
        if len(idxs) == 0:
            return 0
        return int(np.bitwise_xor.reduce(arr[idxs]))
    for start in range(0, n, block_size):
        end = min(n, start + block_size)
        idxs = list(range(start, end))
        a_par = parity_of(alice, idxs)
        b_par = parity_of(bob, idxs)
        if a_par != b_par:
            lo, hi = start, end
            while hi - lo > 1:
                mid = (lo + hi) // 2
                left_idxs = list(range(lo, mid))
                if parity_of(alice, left_idxs) != parity_of(bob, left_idxs):
                    hi = mid
                else:
                    lo = mid
            bob[lo] ^= 1
    return alice, bob

def privacy_amplification(shared_bits, final_key_len, rng):
    n = len(shared_bits)
    if final_key_len <= 0 or final_key_len > n:
```

```
        raise ValueError("final_key_len must be between 1 and length of shared bits")
    R = rng.integers(0, 2, size=(final_key_len, n), dtype=np.uint8)
    out = (R @ shared_bits) % 2
    return out.astype(np.uint8)

def run_bb84_sim(n_bits=1024, p_eve=0.02, channel_error_rate=0.01, sample_fraction=0.05,
block_size=16, final_key_len=128, rng_seed=None):
    rng = np.random.default_rng(rng_seed if rng_seed is not None else secrets.randbits(32))
    alice_bits, alice_bases = prepare_alice(n_bits, rng)
    bob_bases = random_bits(n_bits, rng)
    bob_results = intercept_resend_channel(alice_bits, alice_bases, bob_bases, rng, p_eve=p_eve,
channel_error_rate=channel_error_rate)
    indices, alice_sifted, bob_sifted = sift(alice_bases, bob_bases, alice_bits, bob_results)
    n_sifted = len(indices)
    qber_est, sample_positions, sample_errors = estimate_qber(alice_sifted, bob_sifted,
sample_fraction, rng)
    mask = np.ones(n_sifted, dtype=bool)
    mask[sample_positions] = False
    alice_after_sample = alice_sifted[mask]
    bob_after_sample = bob_sifted[mask]
    alice_corrected, bob_corrected = error_reconciliation_parity(alice_after_sample, bob_after_sample,
block_size, rng)
    matching_mask = alice_corrected == bob_corrected
    shared_bits = alice_corrected[matching_mask]
    success = len(shared_bits) >= final_key_len
    final_key_bits = None
    final_key_hex = None
    if success:
        pa_rng = np.random.default_rng(rng.integers(0, 2**32))
        final_key_bits = privacy_amplification(shared_bits, final_key_len, pa_rng)
        final_key_str = ''.join(str(int(b)) for b in final_key_bits)
        final_key_hex = hex(int(final_key_str, 2))[2:]
    return {
        'alice_bits': alice_bits,
        'alice_bases': alice_bases,
        'bob_bases': bob_bases,
        'bob_results': bob_results,
        'indices': indices,
        'alice_sifted': alice_sifted,
        'bob_sifted': bob_sifted,
        'qber_est': qber_est,
        'sample_positions': sample_positions,
        'sample_errors': sample_errors,
        'shared_bits': shared_bits,
        'final_key_hex': final_key_hex,
        'final_key_bits': final_key_bits,
        'success': success,
        'n_sifted': n_sifted,
    }

class BB84GUI:
    def __init__(self, root):
        self.root = root
        self.root.title("BB84 QKD Visualizer")
        self.style = tb.Style(theme='litera')
        self.mainframe = ttk.Frame(root, padding=10)
        self.mainframe.pack(fill='both', expand=True)
```

```
        self._make_controls()
        self._make_plots()
        self.last_result = None

    def _make_controls(self):
        ctrl = ttk.LabelFrame(self.mainframe, text='Controls', padding=8)
        ctrl.grid(row=0, column=0, sticky='nsew')
        ttk.Label(ctrl, text='Number of qubits:').grid(row=0, column=0, sticky='w')
        self.n_bits_var = tk.IntVar(value=2048)
        ttk.Entry(ctrl, textvariable=self.n_bits_var, width=8).grid(row=0, column=1, sticky='w')
        ttk.Label(ctrl, text='Eve intercept prob:').grid(row=1, column=0, sticky='w')
        self.p_eve_var = tk.DoubleVar(value=0.02)
        ttk.Entry(ctrl, textvariable=self.p_eve_var, width=8).grid(row=1, column=1, sticky='w')
        ttk.Label(ctrl, text='Channel error rate:').grid(row=2, column=0, sticky='w')
        self.chan_err_var = tk.DoubleVar(value=0.01)
        ttk.Entry(ctrl, textvariable=self.chan_err_var, width=8).grid(row=2, column=1, sticky='w')
        ttk.Label(ctrl, text='Sample fraction:').grid(row=3, column=0, sticky='w')
        self.sample_frac_var = tk.DoubleVar(value=0.05)
        ttk.Entry(ctrl, textvariable=self.sample_frac_var, width=8).grid(row=3, column=1, sticky='w')
        ttk.Label(ctrl, text='Final key length (bits):').grid(row=4, column=0, sticky='w')
        self.final_len_var = tk.IntVar(value=256)
        ttk.Entry(ctrl, textvariable=self.final_len_var, width=8).grid(row=4, column=1, sticky='w')

        run_btn = tb.Button(ctrl, text='Run Simulation', bootstyle='success',
command=self.run_simulation)
        run_btn.grid(row=5, column=0, columnspan=2, pady=6)

        save_btn = tb.Button(ctrl, text='Save Key', bootstyle='info', command=self.save_key)
        save_btn.grid(row=6, column=0, columnspan=2, pady=2)

        self.use_qkd_var = tk.BooleanVar(value=True)
        ttk.Checkbutton(ctrl, text='Use QKD key for codon mapping',
variable=self.use_qkd_var).grid(row=7, column=0, columnspan=2, sticky='w', pady=(4,0))

        apply_btn = tb.Button(ctrl, text='Apply QKD Key → Mapping', bootstyle='primary',
command=self.apply_qkd_key)
        apply_btn.grid(row=8, column=0, columnspan=2, pady=(6,0))

        copy_btn = tb.Button(ctrl, text='Copy Key', bootstyle='secondary',
command=self.copy_key_to_clipboard)
        copy_btn.grid(row=9, column=0, columnspan=2, pady=(2,0))

        self.status_lbl = ttk.Label(ctrl, text='Status: idle')
        self.status_lbl.grid(row=10, column=0, columnspan=2, sticky='w', pady=(6,0))

    def apply_qkd_key(self):
        if not self.last_result or not self.last_result.get('success'):
            messagebox.showwarning('No key', 'No final key available. Run simulation first.')
            return
        if not self.use_qkd_var.get():
            messagebox.showinfo('Not enabled', 'The "Use QKD key" checkbox is not checked.')
            return
        hexkey = self.last_result.get('final_key_hex')
        if not hexkey:
            messagebox.showerror('Error', 'No valid key found in last result.')
            return
        try:
```

```
            seed_int = int(hexkey, 16)
        except Exception:
            seed_int = int(hashlib.sha256(hexkey.encode()).hexdigest(), 16)
        codons.set_key(seed=seed_int)
        messagebox.showinfo('Applied', 'QKD key applied to codon mapping (local).')

    def copy_key_to_clipboard(self):
        if not self.last_result or not self.last_result.get('success'):
            messagebox.showwarning('No key', 'No final key available to copy.')
            return
        hexkey = self.last_result.get('final_key_hex')
        if not hexkey:
            messagebox.showerror('Error', 'No valid key found in last result.')
            return
        try:
            self.root.clipboard_clear()
            self.root.clipboard_append(hexkey)
            messagebox.showinfo('Copied', 'QKD key copied to clipboard (hex).')
        except Exception as e:
            messagebox.showerror('Clipboard error', f'Failed to copy to clipboard: {e}')

    def _make_plots(self):
        plot_frame = ttk.LabelFrame(self.mainframe, text='Visualization', padding=8)
        plot_frame.grid(row=0, column=1, sticky='nsew')
        self.fig = Figure(figsize=(8,4))
        self.ax_raster = self.fig.add_subplot(121)
        self.ax_qber = self.fig.add_subplot(122)
        self.ax_raster.set_title('Bits / Bases (raster)')
        self.ax_qber.set_title('QBER (sample)')
        self.fig.tight_layout()
        self.canvas = FigureCanvasTkAgg(self.fig, master=plot_frame)
        self.canvas.get_tk_widget().pack(fill='both', expand=True)

        info_frame = ttk.LabelFrame(self.mainframe, text='Result Info', padding=8)
        info_frame.grid(row=1, column=0, columnspan=2, sticky='nsew', pady=(8,0))
        self.info_text = tk.Text(info_frame, height=8, wrap='word')
        self.info_text.pack(fill='both', expand=True)

    def run_simulation(self):
        self.status_lbl.config(text='Status: running...')
        n_bits = max(64, int(self.n_bits_var.get()))
        p_eve = float(self.p_eve_var.get())
        chan_err = float(self.chan_err_var.get())
        sample_frac = float(self.sample_frac_var.get())
        final_len = int(self.final_len_var.get())
        res = run_bb84_sim(n_bits=n_bits, p_eve=p_eve, channel_error_rate=chan_err,
sample_fraction=sample_frac, final_key_len=final_len)
        self.last_result = res
        self._update_visuals(res)
        self._update_info(res)
        self.status_lbl.config(text='Status: done')

    def _update_visuals(self, res):
        show_n = min(200, len(res['alice_bits']))
        matrix = np.vstack([
            res['alice_bits'][:show_n],
            res['alice_bases'][:show_n],
```

```
            res['bob_bases'][:show_n],
            res['bob_results'][:show_n]
        ])
        self.ax_raster.clear()
        self.ax_raster.imshow(matrix, aspect='auto', interpolation='nearest')
        self.ax_raster.set_yticks([0,1,2,3])
        self.ax_raster.set_yticklabels(['A bits','A bases','B bases','B meas'])
        self.ax_raster.set_xlabel('Position (first %d)' % show_n)

        self.ax_qber.clear()
        q = res['qber_est']
        samp_errs = res['sample_errors']
        samp_n = len(res['sample_positions'])
        self.ax_qber.bar([0], [q*100])
        self.ax_qber.set_ylim(0, max(5, q*100*3))
        self.ax_qber.set_xticks([0])
        self.ax_qber.set_xticklabels([f'QBER sample ({samp_n})'])
        self.ax_qber.set_ylabel('% errors')
        self.fig.tight_layout()
        self.canvas.draw()

    def _update_info(self, res):
        self.info_text.delete('1.0', tk.END)
        lines = []
        lines.append(f"Sifted bits: {res['n_sifted']}")
        lines.append(f"Estimated QBER (sample): {res['qber_est']*100:.3f}% ({res['sample_errors']}
errors in {len(res['sample_positions'])} sampled)")
        lines.append(f"Shared bits available: {len(res['shared_bits'])}")
        if res['success']:
            lines.append(f"FINAL KEY (hex): {res['final_key_hex']}")
        else:
            lines.append("FINAL KEY: <not enough bits to produce requested key length>")
        self.info_text.insert(tk.END, '\n'.join(lines))

    def save_key(self):
        if not self.last_result or not self.last_result.get('success'):
            messagebox.showwarning('No key', 'No final key available to save. Run simulation first and
ensure success.')
            return
        initial = 'qkd_key.txt'
        path = filedialog.asksaveasfilename(defaultextension='.txt', initialfile=initial,
filetypes=[('Text files','*.txt')])
        if not path:
            return
        with open(path, 'w') as f:
            f.write(self.last_result['final_key_hex'])
        messagebox.showinfo('Saved', f'Key saved to: {path}')

if __name__ == "__main__":
    root = tb.Window(themename='litera')
    app = BB84GUI(root)
    root.mainloop()
```

he_demo.py:

Python

```python
from phe import paillier

# Step 1: Generate Paillier keypair
public_key, private_key = paillier.generate_paillier_keypair()

print("Public Key (n):", public_key.n)
print("Private Key (lambda):", private_key.p, private_key.q)

# Step 2: Encrypt two numbers
a = 25
b = 17

enc_a = public_key.encrypt(a)
enc_b = public_key.encrypt(b)

print("\nEncrypted a:", enc_a.ciphertext())
print("Encrypted b:", enc_b.ciphertext())

# Step 3: Homomorphic addition (encrypted)
enc_sum = enc_a + enc_b   # still encrypted

# Step 4: Decrypt the result
result = private_key.decrypt(enc_sum)

print("\nDecrypted result of a + b =", result)
```